



Vectorcardiogram signal compression: A hybrid approach using discrete wavelet transform and singular vector sparse reconstruction

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ABSTRACT

Aims: Vectorcardiography (VCG) records the heart's electrical activity in three dimensions: frontal, transverse, and sagittal for accurate cardiac diagnosis and monitoring. Continuous cardiac monitoring generates vast VCG data daily. The large amount of VCG data poses challenges for storage, dataset management, and transmission, particularly in remote areas with limited resources. The main objective of this research is to develop an efficient VCG data compression technique to streamline the handling and storage of cardiac data, reducing storage space, bandwidth use, and enhancing data transmission speed.

Methods: This work proposes a VCG data compression technique combining Discrete Wavelet Transform (DWT) and Singular Vector Sparse Reconstruction (SVSR) to reduce data size, while preserving vital cardiac information. The process consists of two stages: compression and reconstruction. In compression, the VCG signal undergoes decomposition using the Discrete Wavelet Transform with Haar wavelet, extracting low and high-frequency components. Moreover, the compression efficiency is enhanced by applying SVSR in each subband. In the reconstruction phase, the compressed VCG signal is restored by performing interpolation and the inverse discrete wavelet transform (IDWT), improving overall reconstruction performance.

Results: The results have been tested using PTB diagnostic ECG databases. The effectiveness of the developed compression method is examined quantitatively and qualitatively, in terms of the following metrics; such as compression ratio, signal-to-noise ratio, peak signal-to-noise ratio, percent root mean square difference (PRD) and structural similarity index. According to the performance metrics results, the proposed method achieves 55.67 % higher CR, and 57.12 % improved PRD as compared to existing method.

Conclusions: The proposed method is efficient and adaptable for compressing different VCG signals, offering control over the quality of reconstructed data. Simulated results illustrate that the proposed compression method is a very promising solution for data storage and tele-transmission applications.

Introduction

Nowadays, Cardiovascular disease is now the leading cause of abnormal death worldwide, with cases rising rapidly each year, making it a major global health challenges, according to the World Health Organization [1]. Vectorcardiography (VCG) records the heart's electrical activity in three dimensions such as frontal, transverse, and sagittal for accurate cardiac diagnosis and monitoring [2,3]. VCG signal recorded for continuous cardiac monitoring generates huge amounts of data daily that requires to storage space and transmission bandwidth [4–6]. Therefore, it is essential to compress the VCG data for efficient transmission and storage [7].

Over the years, various compression techniques have been

developed, which are divided into two primary categories: lossless and lossy [7]. Lossless maintains the full signal for precise diagnostics, while lossy reduces data size by eliminating less significant details [8]. In order to achieve high compression with acceptable distortion, many methods for ECG signal have reported in the literature, which focus on lossy compression [9]. These methods are categorized into four groups: direct, linear prediction [10], transform-based [11], and compressed sensing methods [12]. In direct approaches, ECG signals are processed through quantization and coding, excluding transformations. These methods exploit the periodicity and sparsity of the signal, but often yield lower compression efficiency [13,14]. Researchers have introduced different transform based compression techniques such as: discrete wavelet transform (DWT) [15], discrete cosine transform (DCT) [16],

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Beta wavelet [17], and singular value decomposition (SVD) [18]. Transform-based methods are more effective than direct compression approaches, offering higher compression rate and better reconstruction quality. In various studies, 1D ECG signals are converted into 2D representations, where transform-based methods enhance compression. Recently, 2D techniques have significantly improved ECG data compression performance [16]. Commonly, an SVD-based ECG/VCG compression technique has been reported in [19,20], however these methods have limited compression using maximum rank truncation. Additionally, 2D-based compression methods like SVD, DWT, DCT, and ASWDR have been successfully used to compress ECG signals [21]. Literature review evidences that numerous techniques have been explored for the ECG signal processing. However, these methods have limitations, including the need for preprocessing to convert the ECG data into a matrix, which increases the overall complexity. Stephen et al. [22], proposed the VCG data compression using Walsh and Cosine Transforms analysis, and found limited compression. Optimal coding of vectorcardiography sequence using spatial prediction enhances the compression efficiency [23]. However, the effectiveness of spatial prediction for compression may be limited in terms of compression ratio and reconstruction quality. In a recent work [24], authors have developed parameter extraction based linear prediction methods to compress ECG and VCG data. In this approach, architecture may reduce power consumption, but it might not always achieve the highest compression efficiency. In a recent study [25], authors introduced a discrete Karhunen-Loeve expansion and a tunable-quality wavelet transform for VCG data compression. This method offers efficient compression and reconstruction, further it requires careful parameters tuning, and also shows limitations in terms of percent root mean square difference (PRD) and quality score (QS) performance. Based on the literature review, it is seen that only few references are available for the VCG signal compression. The DWT is the most widely used transform for the signal compression due to its superior time-frequency localization capabilities. Authors in [26–28] have employed singular vector sparse reconstruction (SVSR) technique for image compression, which provides good compression and reconstruction quality. Therefore, in above context, a hybrid compression technique is proposed that employs both singular vector sparse reconstruction (SVSR) and discrete wavelet transform (DWT) for compressing the VCG data. The proposed technique effectively compresses the VCG signal, and addresses the limitations associated with VCG compression methods. The goal is to improve compression efficiency, while maintaining reconstruction quality of the VCG signals. The effectiveness of the proposed compression technique is assessed using standard evaluation parameters: such as compression ratio (CR), signal-to-noise ratio (SNR), percent root-mean-square difference (PRD), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM). The contributions of the proposed method are outlined as follows:

1. To improve VCG signal compression, a hybrid compression method based on DWT and SVSR is introduced.
2. SVSR is employed for the first time as a method for compressing VCG signals.
3. Use of sparse sampling eliminates redundancy within the singular vector, improving the overall compression efficiency.
4. Different fidelity metrics are used to examine the effectiveness of the proposed technique.

The structure of the paper is outlined as follows: **Methods** section introduces the proposed compression technique in detail. The experimental setup, and results are included in **Experimental setup and result analysis** section. Discussion and comparison are included in **Discussion** section, and finally the conclusion is presented in **Conclusion** section.

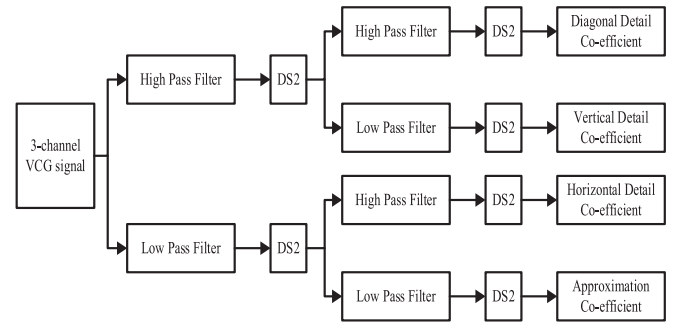


Fig. 1. Schematic of a one-level 2D DWT decomposition.

Methods

This section discusses the proposed methodology used in this paper. VCG signal compression remains an emerging research topic in biomedical, driven by growing applications, including health monitoring and disease detection utilizing VCG signal. This paper proposes a hybrid technique for VCG signal compression based on DWT and SVSR.

Discrete wavelet transform (DWT)

In the field of science and technology, wavelet transform has garnered significant interest, particularly in application of signal and image processing. In 1884, Grossman and Morlet introduced the wavelet concept, which was employed to study signals at different scales, particularly focusing on seismic signals [17]. DWT has gained recognition in signal and image processing for applications like compression, de-noising, enhancement, and classification. DWT has been extensively studied and applied in signal compression due to its ability to provide a multi-resolution analysis, enabling the capture of both high and low-frequency coefficients of a signal [29]. Wavelet transform decomposes a given signal by exploiting the properties of translation and scaling a single prototype function $\varphi(x)$, termed as the mother wavelet, as described in [30],

$$\varphi_{m,n}(x) = |m|^{-1/2} \varphi\left(\frac{x-n}{m}\right), \quad m, n \in \mathbb{R}, \quad m \neq 0, \quad (1)$$

where, m is the parameter for scaling, and n is the translating parameter. If m and n are the constrained such that $m = 2^{-p}$, $n = q2^{-p}$, then emergence of a new wavelet family, called discrete wavelets, expressed as,

$$\varphi_{p,q}(x) = 2^{p/2} \varphi(2^p x - q), \quad p, q \in \mathbb{Z}, \quad (2)$$

The single-level 2D DWT decomposition is shown in Fig. 1.

In DWT, 2-D filter bank structure is used to decompose the VCG data, which contains both low-pass and high-pass filters [31]. After that, the down sampling process is applied in both filters to reduce the number of samples in a signal or image at each level of the decomposition [15]. In DWT decomposition, low-pass and high-pass filters separate the original visual data into discrete frequency bands [29]. The high-pass filter extracts finer details; such as textures, edges, local variances, and within the image data or signal, whereas the low-pass filter captures the approximation details.

Singular vector sparse reconstruction (SVSR)

The SVSR technique based on SVD compresses the image data [26]. SVD is a powerful tool in linear algebra that decomposes a given matrix using Eq. (3), into three distinct matrices, namely: diagonal matrix, orthogonal matrix, and transpose of an orthogonal matrix [32].

$$M_{p \times q} = A_{p \times p} S_{p \times q} B_{q \times q}^T, \quad (3)$$

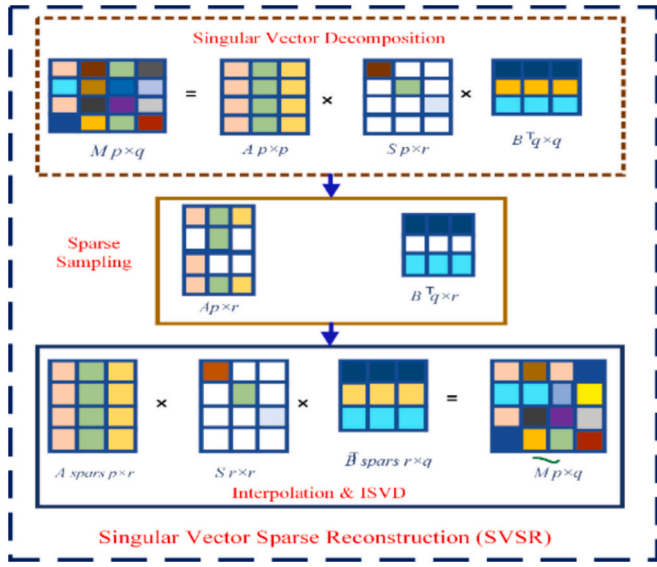


Fig. 2. Framework of SVSR.

where, A and B^T are the orthogonal singular vectors having the size of $p \times r$ and $r \times q$, respectively. $S_{p \times r}$ is the singular value matrix that contains the energy in decreasing order on its main diagonal. The last element of the main diagonal is the smallest singular value that captures the least energy and has least influence on the signal data [33]. The rank of a matrix S is determined by the number of linearly independent rows or columns or equivalently, by the count of non-zero elements on its diagonal. According to [28], these factorized matrices capture the basic information from the original data matrix, such as correlation, amplitude, and important patterns of the signal. Compression can be attained by decreasing the rank value of singular value matrix to approximate the original data matrix. The SVSR block diagram is shown in Fig. 2.

The SVSR technique has improved compression efficiency and reconstruction quality through the utilization of sparse sampling and interpolation.

Proposed method

This section discusses the proposed methodology used in this paper. VCG signal compression remains an emerging research topic in biomedical, driven by the growing applications, including health monitoring and disease detection utilizing VCG signal. This paper proposes a hybrid technique for VCG signal compression based on DWT and SVSR. The proposed method is shown in Fig. 3.

Compression process

In the proposed compression technique, a combination of DWT and SVSR approach has been employed to achieve lossy VCG data compression. In this approach, the first, forward DWT algorithm is applied using low- and high-pass linear filters to decompose the VCG signal into approximation and detailed frequency components, incorporating down-sampling for enhanced algorithm speed. 2D-DWT obtains four subband image coefficients: such as approximation (LL), horizontal (HL), vertical (LH), and diagonal (HH). The approximation subband holds low-frequency components, capturing the VCG signal's key features and preserving most essential information. Meanwhile, high-frequency components are preserved in the detailed subband, often associated with noise or minor fluctuations. The decomposition results of one level are demonstrated in Fig. 4.

In second step, SVSR is applied in all sub bands LL, LH, HL, and HH obtained after DWT decomposition. The SVSR is based on SVD and sparse sampling for compression. In this stage, firstly, the SVD is applied to factorize all the sub band matrices according to Eq. (3), resulting in two orthogonal matrices and a diagonal matrix containing the singular values. Two vectors generate a sparse matrix, using Eq. (4). In this process initially the first and last row of the singular vector matrix is key points. After that, specific coefficients from both the left and right singular vectors are eliminated to improve compression, while preserving the significant key points for good reconstruction quality. The key point is computed using sparse sampling, which is expressed as,

$$\chi_i(\text{key-point}) = \begin{cases} \chi_i & \text{if } ((\chi_i - \chi_{i-1}) \times (\chi_{i+1} - \chi_i) < 0) \\ \chi_i & i = 1 \\ \chi_m & i = m \\ 0 & \text{else} \end{cases} \quad (4)$$

where, $i = \{1, 2, 3, \dots, p\}$, and χ_i depicted key point value. $\chi_i(\text{key-point})$ denotes all the selected key points. The technique ensures that the VCG signal features are maintained, while simultaneously extracting the redundant information from all the sub band (low and high-frequency) coefficients. Since, sparse sampling eliminates some values in singular vectors, those vectors A_i and B_j are generated in a sparse manner. Each unit of a singular vector consists of p discrete values, and these values need to be sampled in a total of r times. After, sparse sampling, the original block size $p \times q$ is replaced with $(p \times fp + q \times fq + 1) \times r$ values. Here, $f(0 < f \leq 1)$, is specified as sampling frequency. After choosing these key points using Eq. (5), the singular vectors A_i and B_j^T are constructed which is a sparse vector matrix, denoted by A_{Isparse} and B_{Isparse}^T . The compressed data is finally acquired from Eq. (5) using the SVSR technique.

$$\tilde{M}_{p \times q} = A_{\text{Isparse}_{p \times r}} S_{r \times r} B_{\text{Isparse}_{r \times q}}^T \quad (5)$$

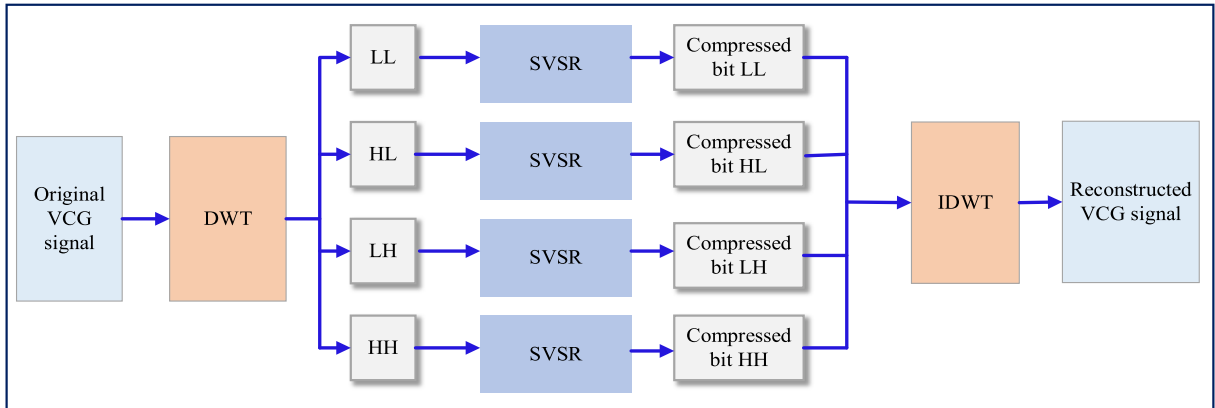


Fig. 3. Proposed methodology for VCG compression.

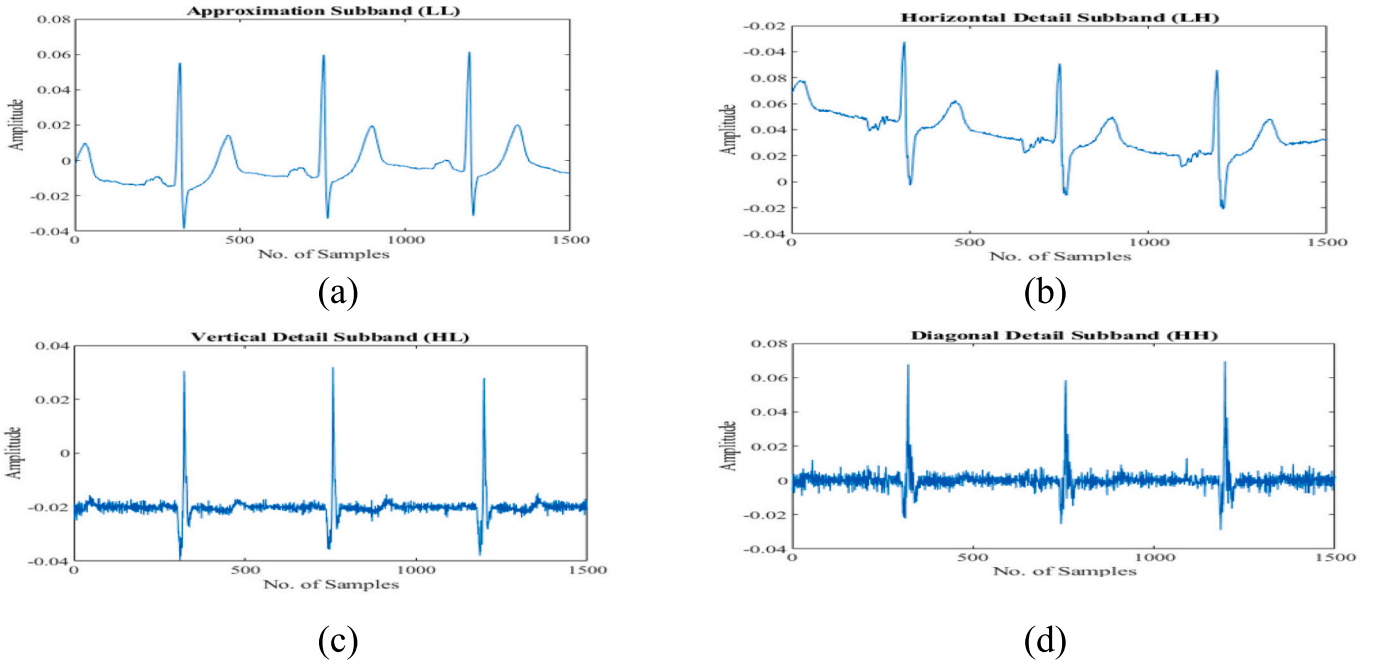


Fig. 4. Approximation and detail coefficients of x lead VCG signal using Haar wavelets.

where, $\tilde{M}_{p \times q}$ is demonstrates as a reconstructed VCG signal matrix with r rank. The SVSR approach utilizes sparse sampling to eliminate the redundant coefficients, achieving significant compression, while preserving the significant information of VCG signal. The next section discusses the reconstruction process.

Reconstruction process

The reconstruction process employs interpolation and inverse DWT to recover the compressed VCG signal. In the first stage of decompression, interpolation is applied in a sparse vector matrix to recover the sparse coefficient in left and right singular vectors. Different interpolation methods are developed for obtaining the missing values or data recovery such as: Piecewise Cubic Hermite Interpolating Polynomial, Spline, Makima, linear and linear interpolation [34], and are utilized for missing data recovery and reconstruction of the sparse matrix. In this study, the authors examine the interpolation approach with proposed technique to efficiently recover the VCG signals. The interpolation can be expressed as:

- **Spline Interpolation (SI):** Spline interpolation is a mathematical technique that estimates the unknown values in the sparse matrix through data points using cubic polynomials with continuous first and second derivatives, which is given in [35],

$$P(x) = p_i + q_i x + r_i x^2 + s_i x^3, \quad (6)$$

- **Piecewise Cubic Hermite Interpolating Polynomial (PCHIP):** A third-degree polynomial, PCHIP is used for interpolating sparse values in a vector. It preserves the shape of data by using information from both the preceding and succeeding neighbouring points, defined in [35],

$$f(x) = p(\chi - \chi_1)^3 + q(\chi - \chi_1)^2 + r(\chi - \chi_1) + s, \quad (7)$$

where, (p, q, r, s) coefficient of the polynomial, in the interval $[\chi_1, \chi_2]$.

- **Makima Interpolation (MI):** It is based on a polynomial interpolation process that approximates a function between two known data

points in the interval $[\chi_i, \chi_{i+1}]$. The method minimizes the undesired fluctuations by selecting derivatives as a linear sequence of nearest slopes (δ_i), mathematically described in [36],

$$w_i = \frac{\delta_1}{\delta_1 + \delta_2} \zeta_{i-1} + \frac{\delta_2}{\delta_1 + \delta_2} \zeta_i \quad (8)$$

where,

$$\delta_1 = \frac{|\zeta_{i+1} - \zeta_i| + |\zeta_{i+1} + \zeta_i|}{2} \quad (9)$$

$$\delta_2 = \frac{|\zeta_{i+1} - \zeta_{i-2}| + |\zeta_{i-1} + \zeta_{i-2}|}{2} \quad (10)$$

where, ζ_i is a slope of the interval $[\chi_i, \chi_{i+1}]$, which is defined as:

$$\zeta_i = \frac{\gamma_{i+1} - \gamma_i}{\chi_{i+1} - \chi_i} \quad (11)$$

- **Linear Interpolation (LI):** This interpolation method is used to estimate the missing values between two known points by applying a straight-line equation, making it a quick and efficient technique for signal and data reconstruction [35]. The process of piece-wise linear interpolation is expressed as:

$$z_i(\chi) = \left(\frac{\chi - \chi_k}{\chi_{k-1} - \chi_k} \right) \times f(\chi_{k-1}) + \left(\frac{\chi - \chi_{k-1}}{\chi_k - \chi_{k-1}} \right) \times f(\chi_k), \quad (12)$$

where, $z_i(\chi)$ represents the value at the point to be interpolated, χ_k denotes the index of the key point $\chi_{k-1} < \chi < \chi_k$, and $f(\chi_k)$ represents the value of the chosen key points $k = (2, 3, 4, \dots, m)$. Further, the compressed subband VCG signal is processed with inverse discrete wavelet transform (IDWT) with the help of proposed technique shown in Fig. 3. The experimental setup, results, and performance evaluation of the proposed method with different interpolation approaches are presents in the next section.

Table 1

Results of the proposed technique using LI and SI interpolation for the VCG dataset.

Data	Linear (LI)						Spline (SI)					
	CR	PSNR	PRD	SSIM	MSE (10^{-3})	Runtime (in s)	CR	PSNR	PRD	SSIM	MSE (10^{-3})	Runtime (in s)
s0001_rem	9.21	23.69	4.22	0.945	4.27	0.51	9.21	18.85	6.44	0.927	13.04	1.06
s0002_rem	9.39	25.56	1.99	0.958	2.78	0.79	9.39	22.43	2.50	0.942	5.71	0.63
s0010_rem	10.98	33.26	2.12	0.982	0.47	0.41	10.98	30.22	2.84	0.977	0.95	0.43
s0020a_rem	12.43	30.81	1.82	0.990	0.83	0.55	12.43	27.23	2.28	0.982	1.89	0.87
s0038l_rem	8.81	29.63	2.12	0.988	1.09	0.74	8.81	28.92	2.25	0.984	1.28	0.99
s0301l_rem	9.70	25.39	3.84	0.964	2.89	0.60	9.70	22.47	5.28	0.950	5.66	0.71
s0304l_rem	9.29	27.06	0.85	0.987	1.97	0.55	9.29	22.16	1.45	0.977	6.08	1.10
s0305l_rem	10.22	23.56	2.40	0.974	4.40	0.48	10.22	20.01	3.17	0.952	9.99	0.60
s0003_rem	9.44	32.09	1.09	0.991	0.62	0.48	9.44	29.83	1.34	0.989	1.04	0.87
s0014l_rem	10.71	28.54	1.75	0.990	1.40	0.46	10.71	24.99	2.54	0.984	3.17	0.93
s0015l_rem	9.30	31.80	2.20	0.990	0.66	0.85	9.30	21.06	6.40	0.944	7.83	0.76
s0016l_rem	9.85	28.52	4.20	0.976	1.41	0.47	9.85	25.06	6.00	0.973	3.12	0.71
s0017l_rem	9.85	30.27	1.72	0.994	0.94	0.45	9.85	28.54	2.23	0.990	1.40	0.57
s0022l_rem	11.21	29.63	1.00	0.999	1.09	0.40	11.21	27.10	1.22	0.997	1.95	0.49
s0025l_rem	10.76	29.54	2.71	0.958	1.11	0.41	10.76	23.46	4.81	0.927	4.50	0.39
s0026l_rem	9.08	25.65	3.88	0.954	2.72	0.52	9.08	24.26	4.63	0.941	3.75	0.39
s0027l_rem	9.18	31.36	1.17	0.997	0.73	0.38	9.18	27.70	1.53	0.990	1.70	0.44
s0028l_rem	9.14	29.42	1.02	0.998	1.14	0.42	9.14	27.77	1.26	0.997	1.67	0.39
s0029l_rem	9.63	24.98	2.00	0.988	3.18	0.40	9.63	23.13	2.31	0.975	4.86	0.43
s0031l_rem	10.19	29.87	1.74	0.994	1.03	0.40	10.19	27.80	2.07	0.986	1.66	0.40
s0021b_rem	10.69	31.41	1.74	0.986	0.72	0.55	10.69	28.61	2.11	0.968	1.38	0.49
s0024b_rem	10.06	24.79	3.20	0.954	3.32	0.46	10.06	18.41	6.00	0.940	1.42	0.66
s0502_rem	10.29	26.85	2.85	0.986	2.07	0.60	10.29	20.03	5.32	0.946	9.94	0.96
s0503_rem	9.35	29.08	3.12	0.978	1.23	0.50	9.35	23.36	5.34	0.949	4.61	0.54
s0505_rem	9.06	29.90	2.42	0.978	1.02	0.93	9.06	27.72	3.05	0.966	1.69	0.46
Average	9.91	28.51	2.29	0.980	1.72	0.53	9.91	24.84	3.37	0.966	4.01	0.65

Experimental setup and result analysis

The experimental setup, evaluation parameters, and results are presented in this section, along with a comparative analysis of the proposed method.

Setup for experiments, dataset information, and evaluation parameters

The proposed VCG data compression method is implemented in MATLAB, and its performance is assessed using a 3.4GHz Intel Core i7 processor with 16 GB RAM. Exploratory data analysis (EDA): The dataset used in this study is sourced from the Physionet ATM, specifically the PTB Diagnostic ECG Database (<https://physionet.org/content/ptbdb/1.0.0>) [37], which includes ECG records from 549 subjects. These subjects include healthy individuals as well as various heart-related conditions. Each dataset comprises 10,000 samples, recorded over 10-second cycles at a sampling rate of 1000 Hz. The VCG data is represented by spatial vectors in 3D coordinate system, with each point on the VCG loop indicating the direction and magnitude of the heart's electrical activity at specific point in time. The X, Y, and Z components of the VCG are recorded as time-series signals that evolve over time, much like traditional ECGs, but in a three-dimensional format. Data acquisition was performed using a non-commercial PTB prototype recorder that supports 16 input channels: 14 for ECGs, 1 for breathing, and 1 for line voltage. The typical 12-lead ECG (I, II, III, aVR, aVL, aVF, V1–V6) and the three Frank leads (Vx, Vy, and Vz) are among the 25 simultaneously recorded signals included in each record [37]. These signals are digitized at 1000 samples per second, and most records contain demographic and medical details. The recorder is equipped with 100 Ω input resistance (DC), ± 300 mV offset compensation, and an input voltage range of ± 16 mV, providing 0–1 kHz bandwidth, synchronous sampling, and 16-bit per sample resolution. It provides cardiac health information for male and female patients in the age group of 17 to 87 years. In this work, the authors decompose the signal at level 1 using DWT and reduce the data size using SVSR with a rank value. The proposed technique performance is assessed using standard evaluation parameters; such as compression ratio (CR), signal-to-noise ratio (SNR), peak signal-to-noise ratio (PSNR), percent root mean square difference

(PRD) and structural similarity index (SSIM). Also, the runtime has been evaluated to examine the method's complexity. These above-mentioned evaluation parameters are described as:

- Compression Ratio (CR):

$$CR = \frac{\text{Original size of input signal}}{\text{Size of compressed signal}} \quad (13)$$

- Percent root mean square difference (PRD in %):

$$PRD = \sqrt{\frac{\sum [A(n) - \tilde{A}(n)]^2}{\sum A(n)^2}} \quad (14)$$

- Signal-to-noise ratio (SNR):

$$SNR = 10 \log_{10} \left(\frac{V_{\text{signal}}}{V_{\text{noise}}} \right) \quad (15)$$

- Peak signal-to-noise ratio (PSNR) in (dB):

$$PSNR = 10 \log_{10} \left(\frac{\text{peak probable value}^2}{\frac{1}{pq} \sum_{i=1}^p \sum_{j=1}^q [A(i,j) - \tilde{A}(i,j)]^2} \right) \quad (16)$$

- Structural similarity index measure (SSIM):

$$SSIM = \frac{(2\alpha_A \alpha_{\tilde{A}} + c_1)(2\sigma_{A\tilde{A}} + c_2)}{(\alpha_A^2 \alpha_{\tilde{A}}^2 + c_1)(\sigma_A^2 + \sigma_{\tilde{A}}^2 + c_2)} \quad (17)$$

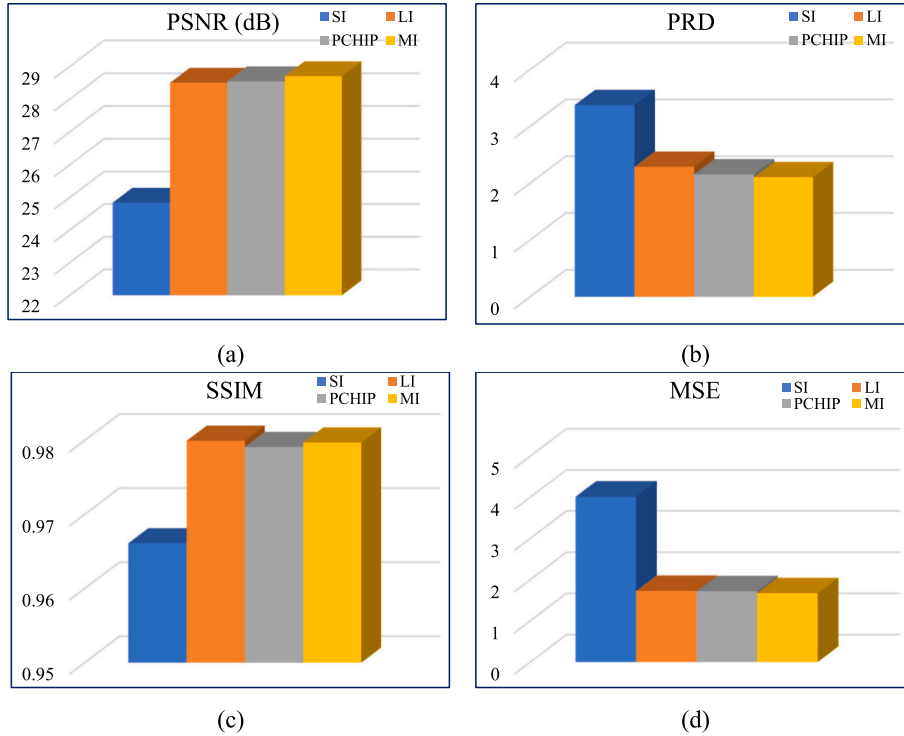
- Quality Score (QS):

$$\frac{CR}{PRD} \quad (18)$$

Table 2

Results of the proposed technique using PCHIP and MI interpolation for the VCG dataset.

Data	PCHIP						Makima (MI)					
	CR	PSNR	PRD	SSIM	MSE (10^{-3})	Runtime (in s)	CR	PSNR	PRD	SSIM	MSE (10^{-3})	Runtime (in s)
s0001_rem	9.21	23.77	3.82	0.950	4.19	0.56	9.21	23.90	3.74	0.950	4.07	0.41
s0002_rem	9.39	26.34	1.66	0.958	2.32	0.74	9.39	26.34	1.64	0.958	2.32	1.06
s0010_rem	10.98	33.53	1.94	0.986	0.44	0.46	10.98	33.89	1.86	0.987	0.41	0.56
s0020arem	12.43	30.13	1.78	0.991	0.97	0.65	12.43	30.06	1.79	0.991	0.99	0.72
s0038lrem	8.81	29.86	2.00	0.985	1.03	0.77	8.81	29.67	2.03	0.985	1.08	0.36
s0301lrem	9.70	25.49	3.52	0.964	2.82	0.99	9.70	25.58	3.47	0.965	2.77	0.58
s0304lrem	9.29	27.74	0.78	0.992	1.68	0.50	9.29	28.05	0.74	0.993	1.57	0.43
s0305lrem	10.22	23.23	2.16	0.971	4.75	0.44	10.22	23.33	2.13	0.972	4.65	1.08
s0003_rem	9.44	33.18	0.95	0.994	0.48	0.60	9.44	34.05	0.87	0.994	0.39	0.93
s0014lrem	10.71	29.45	1.64	0.989	1.14	0.43	10.71	29.53	1.62	0.989	1.11	0.49
s0015lrem	9.30	31.36	2.14	0.990	0.73	2.14	9.30	31.52	2.04	0.991	0.70	0.52
s0016lrem	9.85	29.51	3.85	0.986	1.12	3.85	9.85	30.05	3.67	0.988	0.99	1.29
s0017lrem	9.85	30.57	1.65	0.993	0.88	0.81	9.85	30.66	1.60	0.993	0.86	0.62
s0022lrem	11.21	29.53	0.93	0.998	1.11	0.52	11.21	29.54	0.94	0.998	1.11	0.67
s0025lrem	10.76	28.42	2.88	0.947	1.44	0.50	10.76	28.57	2.82	0.948	1.39	0.43
s0026lrem	9.08	25.41	3.85	0.949	2.88	0.51	9.08	25.37	3.80	0.950	2.90	0.44
s0027lrem	9.18	31.45	1.10	0.996	0.72	0.53	9.18	31.44	1.09	0.996	0.72	0.47
s0028lrem	9.14	29.38	1.00	0.998	1.15	0.46	9.14	29.44	0.98	0.998	1.14	0.36
s0029lrem	9.63	24.87	1.93	0.980	3.26	0.47	9.63	25.09	1.88	0.981	3.10	0.42
s0031lrem	10.19	29.54	1.64	0.989	1.11	0.51	10.19	29.68	1.60	0.989	1.08	0.36
s0021brem	10.69	30.59	1.59	0.974	0.87	0.50	10.69	30.74	1.56	0.975	0.84	0.62
s0024brem	10.06	24.93	3.02	0.964	3.21	1.12	10.06	25.01	2.95	0.966	3.16	0.86
s0502_rem	10.29	26.28	2.74	0.984	2.35	0.40	10.29	26.37	2.73	0.984	2.31	0.70
s0503_rem	9.35	29.49	2.89	0.977	1.12	0.40	9.35	29.84	2.73	0.978	1.04	0.56
s0505_rem	9.06	29.70	2.36	0.973	1.07	0.40	9.06	29.90	2.28	0.974	1.02	0.95
Average	9.91	28.55	2.15	0.979	1.71	0.77	9.91	28.70	2.10	0.980	1.67	0.64

**Fig. 5.** Comparative analysis of the interpolations in the proposed method using (a) PSNR, (b) PRD, (c) SSIM, and (d) MSE for VCG signals.

Results and performance analysis of proposed technique with different interpolations

This section presents the simulation results and their analysis, evaluating various interpolation approaches for VCG signal compression using the proposed technique. The experimental results obtained by the proposed method with different VCG datasets using Haar wavelet filter and different interpolations are presents in [Tables 1 and 2](#).

It is observed from [Tables 1 and 2](#) that LI, SI, MI, and PCHIP interpolation produce similar CR values, on an average of 9.83 at rank 2. It is also evident from these tables that all the interpolation achieves desirable outcomes in terms of PSNR, PRD, SSIM, and MSE for an equal number of VCG data sets. Among the interpolation methods, Makima shows superior average performance based on PSNR (28.70), PRD (2.10), and MSE (1.67).

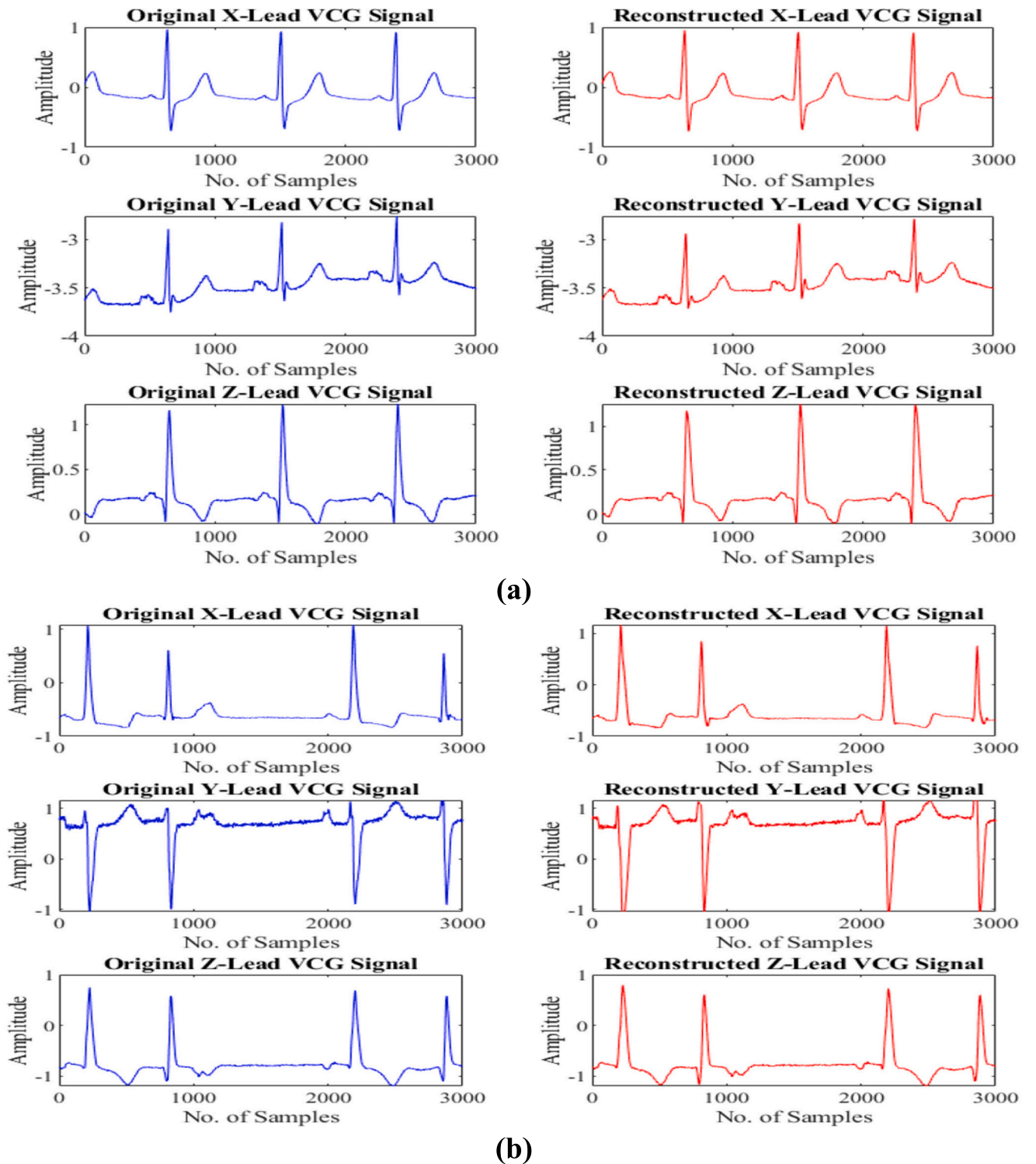


Fig. 6. Original and reconstructed VCG signals using the proposed method with MI interpolation for the datasets (a) s0304lrem and (b) s0024brem.

Discussion

Fidelity assessments

The simulated results obtained using the proposed technique with Haar wavelet filter and different interpolation methods at different VCG signals are reported in Tables 1 and 2. From Tables 1 and 2, the LI, MI, and PCHIP interpolation achieves an SSIM value for all the VCG signals is approximately 0.980, whereas the SI interpolation results in a lower

average SSIM value of 0.962 across the VCG datasets. The computational time requires for the proposed method using different interpolation approaches is summarized in Tables 1 and 2. Therefore, the proposed method significantly requires lower runtime for each VCG dataset on average, with times of 0.53, 0.65, 0.64, and 0.77 s for LI, SI, MI, and PCHIP interpolation, respectively. It can be concluded that use of MI approach in the proposed technique may yield better reconstruction quality than other interpolation methods.

The comparison of the proposed technique with different

Table 3

The effectiveness of the proposed method is compared with the method in [24].

VCG Data	Method in [24]		Propose method using MI interpolation	
	CR	PRD	CR	PSNR
s0014lrem	4.4224	1.5758	10.7100	1.6200
s0017lrem	4.4442	2.8174	9.8500	1.6000
s0020are	4.4594	5.0103	12.4300	1.7900
s0021brem	4.4426	7.5059	10.6900	1.5600
s0022lrem	4.4764	1.7344	11.2100	0.9400
s0025lrem	4.4418	3.8743	10.7600	2.8200
s0026lrem	4.4441	7.1845	9.0800	3.8000
s0027lrem	4.4713	3.9453	9.1800	1.0900
s0028lrem	4.4379	1.5809	9.1400	0.9800
s0029lrem	4.4382	7.2152	9.6300	1.8800
s0031lrem	4.4382	7.2152	10.1900	1.6000
s0038lrem	4.4807	5.3500	8.8100	2.0300
s0301lrem	4.4382	7.2152	9.7000	3.4700
s0015lrem	4.4281	5.0764	9.3000	2.0400
s0016lrem	4.4258	4.7334	9.8500	3.6700
Average	4.4460	4.8023	10.0353	2.0593

interpolation method in terms of PSNR, PRD, SSIM, and MSE is depicted in Fig. 5. The proposed algorithm with MI interpolation provides maximum PSNR (28.70), PRD (2.10), and MSE (1.67) from Figs (a), (b), and (d), respectively. The proposed compression technique with MI interpolation provides 13.45 %, 0.69 %, and 0.54 % higher PSNR, 55.64 %, 8.06 %, and 2.34 % higher PRD, and 58.40 %, 3.18 %, and 2.68 % higher MSE as compared to SI, LI, and PCHIP interpolation respectively. Similarly, it can also be seen from Fig. 5(c) that the proposed

compression technique provides nearly equal SSIM.

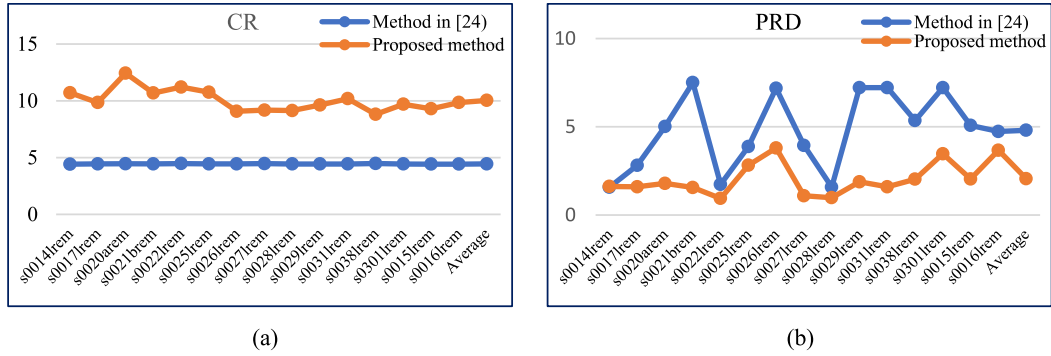
Fig. 6 show the original and reconstructed samples of PTB VCG signals obtained using the proposed technique with MI interpolation: (a) s0304lrem and (b) s0024brem. It is observed that the reconstructed signal exhibits a striking similarity to the original VCG signal. However, the effectiveness of the proposed method has been assessed through testing on different VCG datasets. Therefore, the simulation findings indicate that the proposed technique is suitable and effective for compressing VCG signals.

Performance comparison of the proposed with existing method

The propose method performance is compared to existing methods in [24], and the simulation results for 15 VCG records are presented in Table 3. It can be seen that the developed method achieved 55.67 % higher CR and 57.12 % better PRD as compared to the existing method [24].

Based on the performance analysis of the proposed technique, it is observed that its performance depends on the datasets. The efficiency of the proposed technique on different VCG data sets has also been compared in terms of CR and PRD, as shown in Fig. 7.

This analysis shows that the proposed method using MI approach gives higher signal compression with excellent reconstruction quality in terms of PRD for all the VCG data sets compared to method in [24]. Consequently, it can be said that the proposed compression technique would work incredibly well for both VCG signal compression and reconstruction. The compression performance of the proposed technique has been assessed in terms of reconstruction quality for various VCG

**Fig. 7.** The proposed method is compared with the existing method based on (a) CR and (b) PRD.**Table 4**

Comparative analysis with current approaches.

Methods	Parameters	s0305lrem	s0306lrem	s0320lrem	s0350lrem	Average
SVD [20]	CR	3.74	3.92	2.57	5.36	3.89
	PRD	7.49	7.53	2.63	4.05	5.42
	PSNR	36.92	32.97	41.72	41.01	38.15
	QS	0.49	0.52	0.97	1.32	0.82
DCT [25]	CR	7.35	8.82	11.73	12.2	10.03
	PRD	10.38	10.74	14.56	12.2	11.97
	PSNR	33.83	32.19	32.88	35.02	33.48
	QS	0.71	0.82	0.81	1.05	0.85
KLT [25]	CR	15	15	15	15	15
	PRD	16.62	13.39	25.39	18.11	18.38
	PSNR	33.89	38.8	31.3	31.77	33.94
	QS	0.9	1.12	0.59	0.83	0.86
TQWT [25]	CR	15.72	16.59	20.92	12.76	16.5
	PRD	9.58	8.16	8.16	7.36	8.32
	PSNR	34.53	34.58	37.91	39.41	36.61
	QS	1.64	2.03	2.56	1.73	1.99
Proposed method	CR	10.22	9.98	8.8	10.69	9.92
	PRD	2.12	2.61	0.85	2.37	1.99
	PSNR	23.33	21.65	29.64	29.01	25.91
	QS	4.8	3.83	10.35	4.51	5.87

data sets. The analysis clearly shows that the proposed method is very effective for compression and reconstruction of various VCG signals, outperforming the other existing techniques in terms of efficiency and quality. The proposed technique has also been compared with the methods given in [20,25]. The results in terms of fidelity parameters for different VCG signals, namely s0305lrem, s0306lrem, s0320lrem, and s0350lrem, are summarized in Table 4. Results in Table 4 indicate the superiority of the proposed technique over the SVD compression technique [20], and DCT, KLT, and TQWT techniques [25] in terms of Percent root mean square difference (PRD) and Quality Score (QS) results. Whereas, a slightly low CR has been obtained as compared to KLT and TQWT based compression techniques. A high QS indicates efficient compression with minimal loss of clinically relevant information. A higher QS and lower PRD values reflect a more effective VCG compression algorithm.

The proposed VCG compression technique can improve the storage and transmission of VCG data in clinical settings, making it easier for healthcare providers to manage large volumes of patient data, while maintaining diagnostic accuracy. This is especially important in resource-limited environments, such as telemedicine and remote monitoring, where reducing bandwidth usage and storage requirements is critical. The ability to transmit compressed VCG data efficiently can facilitate faster, more effective remote patient monitoring, enabling timely interventions. However, based on PRD and QS results show that the proposed method maintains reconstruction quality, supporting accurate diagnosis and long-term patient monitoring. These revisions aim to make the practical benefits of our method more explicit and clinically grounded.

Conclusion

This paper presents a new technique that combines DWT and SVSR for effective VCG signal compression and reconstruction. MATLAB is used to implement the proposed compression method, and its performance is analysed using 15 various VCG signals. The input VCG signal is decomposed using Haar wavelet and four sub bands are obtained. Moreover, SVSR is applied across all the sub bands to increase the compression efficiency using sparse sampling. Finally, the interpolation method and inverse discrete wavelet transform (IDWT) are applied to reconstruct the VCG signal. The effectiveness of the proposed method is tested on the PTB diagnostic ECG databases. Its performance is evaluated both quantitatively and qualitatively, using parameters; such as compression ratio (CR), percent root mean square difference (PRD), peak signal-to-noise ratio PSNR, signal-to-noise ratio SNR, mean square error (MSE), and structural similarity index (SSIM). The proposed algorithm with MI interpolation, provides maximum PSNR (28.70), PRD (2.10), and MSE (1.67). The proposed compression technique with MI interpolation provides 13.45 %, 0.69 %, and 0.54 % higher PSNR, 55.64 %, 8.06 %, and 2.34 %, higher PRD, and 58.40 %, 3.18 % and 2.68 % higher MSE as compared to SI, LI, and PCHIP interpolation, respectively. The analysis of the performance metrics reveals a 55.67 % increase in CR and a 57.12 % reduction in PRD, outperforming the existing method.

CRedit authorship contribution statement

Deepak Mishra: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anil Kumar:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

There is no conflict of interest between any of the authors.

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